

Linear systems and elimination

Sharp row-deletion singular-value limit for spherical random matrices

Difficulty: challenging **Importance:** interesting to the community

Status: [Lean verified](#)

Independently reviewed resolution - 2026-09-11

Affirmative resolution. Theorem 1 and Sections 2-5 prove the displayed Gaussian trimmed-second-moment limit in probability along every sequence $n \rightarrow \infty$ and $m/n \rightarrow \infty$, with exactly $\lfloor \theta m \rfloor$ retained rows and the variational least singular value. Explicit failure-probability and error bounds are included; no faster aspect-ratio growth assumption is added.

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The difficulty, importance and rating rationale below are historical assessments of the original open target. Original statements, references and dated audits are preserved.

Rating rationale: Challenging reflects a sharp limit for the least singular value after adversarial row deletion; community impact includes random matrix theory and robust iterative solvers.

Lean proof and verification evidence - 2026-09-22

The complete original target is Lean verified. The [formal proof at revision 1eb284b](#) proves the stated probability limit for every sequence with diverging dimension and aspect ratio. It includes the explicit finite-size failure and error bounds, the exact floor convention, arbitrary changing probability spaces, and the actual normalized surface distribution. The Euclidean operator norm and attained row-deletion minimum are proved to have their original meanings, including rank-deficient and empty-retention cases.

Lean formalization: George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, with substantial AI-agent assistance. **Matthew J. Colbrook** retains attribution for the original mathematical proof. The original statement, source attribution and dated history below remain preserved.

All [23 required declarations](#) passed authentic LeanCert kernel-trust assertions, the sandboxed Lean4 Comparator and kernel replay on non-root Linux. Each transitive axiom closure contains only *propext*, *Classical.choice* and *Quot.sound*. The exact analytic proof uses no approximate numerical integration. The [operational evidence](#) records immutable inputs, authenticated dependency revisions and successful isolation and rejection controls.

Two nonauthor agents independently reviewed the complete statements and proofs: [fidelity review](#) and [correctness review](#). Their [fidelity completion review](#) and [correctness completion review](#) separately check the Linux evidence. Publication correspondence is reviewed in subsequent addenda. These are independent AI reviews under the repository's adapted Tau Ceti protocol, not external human peer review or official Tau Ceti endorsement. See the [project guide](#) and [formalization metadata](#) for exact scope, review history and reproduction instructions.

Problem statement

Fix $0 < \theta < 1$. For $A \in \mathbb{R}^{m \times n}$ define

$$s_\theta(A) = \min_{S \subseteq \{1, \dots, m\}, |S| = \lfloor \theta m \rfloor} \min_{\|x\|_2=1} \|A_S x\|_2,$$

where A_S contains the rows indexed by S . Let $a_\theta > 0$ satisfy $\mathbb{P}(|G| \leq a_\theta) = \theta$ for $G \sim N(0, 1)$, and set

$$h_\theta = \frac{1}{\sqrt{2\pi}} \int_{-a_\theta}^{a_\theta} t^2 e^{-t^2/2} dt.$$

For every integer sequence (m_j, n_j) with $n_j \rightarrow \infty$ and $m_j/n_j \rightarrow \infty$, let the rows of $A_j \in \mathbb{R}^{m_j \times n_j}$ be independent and uniformly distributed on $\mathbb{S}^{n_j-1}$. Is

$$\frac{s_\theta(A_j)^2}{\|A_j\|_2^2} \xrightarrow{\mathbb{P}} h_\theta?$$

This states Steinerberger's proposed asymptotic with $\theta = q - \beta$. Convergence in probability and the floor convention make the source's limiting statement explicit. The source also asks for quantitative error bounds; these belong to this problem rather than separate entries.

Connection to numerical linear algebra

The quantity measures the worst conditioning remaining after rows are discarded. It controls convergence guarantees for quantile Kaczmarz methods on corrupted linear systems.

References

1. S. Steinerberger, *Quantile-based Random Kaczmarz for corrupted linear systems of equations*, Information and Inference **12**(1) (2023), 448–465, §2.3, equation (8) and the following paragraph. [Published paper](#). [Author preprint](#), v1 (2021), §2.3, equation ($\diamond$).
2. E. Battaglia, J.-F. Cai, J. Chen, A. Ma, D. Needell, and T. Wu, *Quantile Randomized Kaczmarz for Streaming Linear Systems with Massart Noise*, arXiv:2608.27968v1 (2026), §1.1, discussion of Steinerberger’s random-matrix heuristic and the distinction between static and streaming data; §5. [Preprint](#).

Earlier status check — 2026-09-08

Checked the published formulation and the arXiv version history (2107.05554 has only v1), then searched the problem’s title, the author’s name with “subsingular”, and combinations of “trimmed/smallest singular value”, “quantile”, “Kaczmarz”, “heuristic”, “proof”, “2025”, and “2026”. The August 2026 streaming paper still describes the original estimate as a heuristic; its fresh-sample convergence bounds do not establish this static-matrix limit. No later proof or counterexample was located in this bounded search. That is a literature-status assessment, not a proof of openness.

Audit update — 2026-09-10

Rechecked Steinerberger’s [primary formulation](#), §2.3, and the [2026 streaming follow-up](#). Searches found no proof of the static proportional-deletion limit. Cai, Chen, Ma, and Wu’s new [subsample-size theorem](#), SIAM J. Matrix Anal. Appl. **47** (2026), 802–823, concerns quantile estimation during QRK, a different asymptotic quantity.